

Functional genetic report · longevity

# Sexual and Erectile Function

Sérgio F · Report date: 02-10-2026

What the genes say about erectile function: the neural part (hypothalamus), the vascular part (nitric oxide, coronary disease, diabetes) and the hormonal part, plus the way drugs are already taken are metabolized. Only genes (chip + local imputation + polygenic scores).

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## Summary

**11**

SNPs analyzed

**6**

in the chip

**5**

imputed ( $DR2 \geq 0.8$ )

**0**

SNP not found in the data

**5**

Polygenic scores

The erection depends on three systems: the brain (sexual excitation, hypothalamus), the vessels (nitric oxide released by the endothelium and nerves of the penis, which activates guanylate-cyclase and produces cGMP) and hormones (testosterone). The tadalafil acts in the last step, because it prevents the PDE5 from degrading the cGMP. The GWAS of erectile dysfunction found a strong locus, SIM1, which acts in the hypothalamus and not in the vessels, and several loci shared with coronary disease and diabetes.

In synthesis: favourable genetic profile. There is no risk allele in SIM1, the strongest locus, nor in PHF21B, and the erectile dysfunction score is at the 15th percentile. The overall coronary load is low (coronary consensus at the 18th percentile; metaGRS single score at the 5th) and genetic testosterone is high (the 89th percentile). The less good side is endothelial nitric oxide: it is heterozygous in the two classical variants of NOS3 (Glu298Asp and -786T>C), a small effect that tadalafil and telmisartan already taking compensates directly.

### Favourable

- **SIM1 without risk alleles** (rs17185536 CC and rs57989773 TT): ~24% of the reference population alleles are at risk in this locus, the strongest of erectile dysfunction (OR ~1.25 per allele).
- **PHF21B rs8141413 CC and SLC39A8 AA:** without risk alleles. Erectile dysfunction score (3 loci) at the 15th percentile.
- **Vascular and hormonal components:** coronary disease at the 18th percentile (consensus; metaGRS single score at the 5th); total genetic testosterone at the 89th percentile (SHBG at the 87th percentile).

### Unfavourable

- **NOS3 Glu298Asp (TG) and -786T>C (CT) heterozygotes:** less endothelial nitric oxide (small effect, per copy).
- **ZC3HC1 GA:** a locus allele shared between erectile dysfunction and coronary artery disease (typical of the population).
- **CYP3A5 \*1/\*3 (rs776746 CT):** expressed functional CYP3A5, less common (~10% of people); it can quickly eliminate some drugs from CYP3A. It is information, not judgment.

**Interpretation:** the read genetics does not point to erectile dysfunction. The genetic weak point (endothelium nitric oxide) is exactly what tadalafil reinforces. The most important factors are modifiable and are already well covered in the plan: physical activity, insulin sensitivity, low LDL and ApoB and absence of alcohol.

## How to read this report

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**Genetics only.** All conclusions come only from Sérgio F genes (without blood tests or exams). They are predispositions, non-diagnosis. Each theme is divided into favourable and unfavourable; what does not have clear judgment appears as typical.

**Strand:** All genotypes are in the strand + of dbSNP (GRCh38). In the SNPs of this report, 6 directly agree with the dbSNP alleles, 0 needed conversion to the complementary strand and 0 are palindromic (A/T or C/G; orientation confirmed by the GRCh38 position). The genotypes are already in the strand + of the reference. Therefore, some genotypes appear with different letters from those used in the literature when using the strand of the gene (e.g. MTHFR C677T = rs1801133 A in the strand +).

**Origin and trust:** chip = directly genotyped (high confidence); imputation DR2=x = locally inferred from a public panel of complete genomes (30x), accepted only with  $DR2 \geq 0.8$  (good trust, just below the chip); SNP not found in the data : neither chip nor imputed with sufficient quality (indels, rare variants, deletions).

**Effect and judgment allele:** The comparison between the number of Sérgio F copies and the mean of the reference population ( $2 \times$  allele frequency; column «Freq.»). **Unfavourable** = at least 0.6 copies of risk allele above average (or any copy of a rare variant of strong effect); **Favourable** = at least 0.6 below average, protective allele, or absence of a strong effect variant; **Typical / no judgement** = close to the average (e.g. homozygous for a risk allele that is the most common) or context variant. Each common SNP has a small individual effect; what weighs more are the variants of large effect (marked as strong evidence) and the polygenic scores.

**Polygenic scores (PGS):** add thousands of variants with coefficients published in PGS Catalog. The percentile compares Sérgio F with the reference population (n=633), calculated with exactly the same variants. Percentage  $\geq 80$  in a risk score = unfavourable;  $\leq 20$  = favourable.

## Polygenic scores

The erectile dysfunction score uses only the 3 loci of the GWAS Catalog with available effect size; it is at the 15th percentile (average percentile, because with few loci there are many draws;  $z=-1.07$ ). Coronary metaGRS and GPS of diabetes come from complete imputation. The testosterone and SHBG scores were calculated in the andropause report. There is no published score for erectile dysfunction in PGS Catalog.

| Trait                                             | Score       | Variables used (coverage)    | Percentage (reference population) | Judgement              |
|---------------------------------------------------|-------------|------------------------------|-----------------------------------|------------------------|
| Erectile dysfunction (GWAS Catalog score, 3 loci) | C_DISF_ERET | 3 of 3 (100%)                | <b>p14</b>                        | Favourable             |
| Coronary disease (metaGRS, 1.7 M variants)        | PGS000018   | 1 476 469 of 1 745 179 (85%) | <b>p5</b>                         | Favourable             |
| 2 type diabetes (GPS, 6.9 M variants)             | PGS000014   | 6 393 464 of 6 917 436 (92%) | <b>p57</b>                        | Typical / no judgement |
| Total testosterone in man                         | PGS003559   | 972 627 of 979 739 (99%)     | <b>p89</b>                        | Favourable             |
| SHBG                                              | PGS003552   | 972 627 of 979 739 (99%)     | <b>p87</b>                        | Typical / no judgement |

# Loci of erectile dysfunction (GWAS)

The GWAS of erectile dysfunction in large biobanks (more than 220 000 men) found the locus SIM1, which regulates the circuits of the hypothalamus involved in sexual excitation and satiety, and several other smaller loci.

**Polygenic scores of this theme:** Erectile dysfunction (GWAS Catalog score, 3 loci) (C\_DISF\_ERET): the 14th percentile Favourable

| Gene                  | SNP        | Genotype (+) | Source                 | Effect allele | Copies | Freq. | Effect                                                                                                                                                                        | Judgement              |
|-----------------------|------------|--------------|------------------------|---------------|--------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| SIM1 locus 6q16       | rs17185536 | CC           | imputation<br>DR2=0.94 | T             | 0      | 0.24  | Stronger locus of erectile dysfunction (OR ~1.25 by T allele), acts in the hypothalamus (sexual excitation circuits), regardless of vascular factors. (strong evidence)       | Favourable             |
| SIM1 secondary signal | rs57989773 | TT           | imputation<br>DR2=0.91 | C             | 0      | 0.24  | According to SNP of the same locus SIM1 (OR ~1.2 by allele C); in linkage unbalance with the previous one, it is not counted separately. (moderate evidence)                  | Typical / no judgement |
| PHF21B                | rs8141413  | CC           | imputation<br>DR2=0.96 | T             | 0      | 0.34  | Locus of erectile dysfunction in large biobank (T allele increases the risk). (moderate evidence)                                                                             | Favourable             |
| SLC39A8               | rs13135092 | AA           | chip                   | G             | 0      | 0.08  | Manganese and zinc transporter ZIP8; G allele (less common) increases the risk of erectile dysfunction and binds to several vascular and cerebral traits. (moderate evidence) | Typical / no judgement |

|                                                                                                                                                                                         |                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>Favourable</b> <ul style="list-style-type: none"><li>• SIM1 rs17185536 CC and rs57989773 TT (0 risk alleles)</li><li>• PHF21B rs8141413 CC</li><li>• SLC39A8 rs13135092 AA</li></ul> | <b>Unfavourable</b> <ul style="list-style-type: none"><li>• None</li></ul> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|

**Interpretation:** Neural/central favourable profile: the 3 loci score is at the 15th percentile. The effect of SIM1 does not depend on vessels, so it is a different information from nitric oxide.

# Vascular component: endothelial nitric oxide and loci shared with coronary disease

Erectile dysfunction is often the first sign of vascular disease, because the arteries of the penis are small (1–2 mm) and suffer before coronary. Endothelial nitric oxide (NOS3) and the burden of coronary risk and diabetes are vascular genetic factors.

**Polygenic scores of this theme:** Coronary disease (metaGRS, 1.7 M variants) (PGS000018): the 5th percentile (single score; coronary consensus p18) **Favourable**; 2 type diabetes (GPS, 6.9 M variants) (PGS000014): the 57th percentile **Typical / no judgement**

| Gene              | SNP        | Genotype (+) | Source                 | Effect allele | Copies | Freq. | Effect                                                                                                                                                                                                | Judgement              |
|-------------------|------------|--------------|------------------------|---------------|--------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| ZC3HC1            | rs56179563 | GA           | imputation<br>DR2=0.95 | G             | 1      | 0.61  | Shared locus between erectile dysfunction and coronary artery disease (G allele). (weak evidence)                                                                                                     | Typical / no judgement |
| NOS3<br>Glu298Asp | rs1799983  | TG           | imputation<br>DR2=1.00 | T             | 1      | 0.35  | Endothelial nitric oxide: the Asp298 form (T allele) is associated in meta-analyses with more erectile dysfunction and less vascular response. Detail in the nitric oxide report. (moderate evidence) | Typical / no judgement |
| NOS3<br>-786T>C   | rs2070744  | CT           | chip                   | C             | 1      | 0.44  | NOS3 promoter: C allele reduces transcription of the enzyme; associated with erectile dysfunction in several meta-analyses. (moderate evidence)                                                       | Typical / no judgement |

Favourable

- Low coronary risk: consensus at the 18th percentile (metaGRS single score at the 5th)
- NOS3 rs3918226 without the T allele (Nitric oxide report)

Unfavourable

- NOS3 Glu298Asp TG
- NOS3 -786T>C CT
- ZC3HC1 GA (typical)
- 2 (GPS) type diabetes in the 57 (medium) percentile

**Interpretation:** Overall good vascular component: low genetic coronary risk. The two variants of NOS3 in heterozygous reduce a little the production of nitric oxide; the tadalafil potentiates precisely the signal of nitric oxide (cGMP). The genetic risk of diabetes is medium and controllable (dapagliflozin and retatrutide already in the plan).

# Testosterone and SHBG

Testosterone is necessary for sexual desire and for the expression of neuronal penis NO- synthase. The testosterone and SHBG scores were calculated in the andropause report and are reused here.

**Polygenic scores of this theme:** Total testosterone in man (PGS003559): the 89th percentile Favourable; SHBG (PGS003552): the 87th percentile Typical / no judgement

| Gene | SNP        | Genotype (+) | Source | Effect allele | Copies | Freq. | Effect                                                                                                                               | Judgement              |
|------|------------|--------------|--------|---------------|--------|-------|--------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| SHBG | rs12150660 | GT           | chip   | T             | 1      | 0.24  | The T allele increases SHBG and total testosterone (free changes little). Seen in detail in the andropause report. (strong evidence) | Typical / no judgement |

**Favourable**

- Total genetic testosterone at the 89th percentile
- SHBG rs12150660 GT (plus SHBG and total testosterone)

**Unfavourable**

- None

**Interpretation:** Genetic profile favourable. On the side of drugs, rapamycin can lower testosterone in some men; isotretinoin was associated in case reports with sexual dysfunction.



# Pharmacogenetics of tadalafil and telmisartan

The tadalafil is mainly eliminated by CYP3A4. Telmisartan blocks the AT1 receptor of angiotensin II and is eliminated by glucuronidation, without depending on CYP3A.

| Gene         | SNP        | Genotype (+) | Source | Effect allele | Copies | Freq. | Effect                                                                                                                                                  | Judgement              |
|--------------|------------|--------------|--------|---------------|--------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| CYP3A4 *22   | rs35599367 | GG           | chip   | A             | 0      | 0.05  | CYP3A4 eliminates tadalafil; the allele *22 (A) reduces the enzyme and increases drug exposure. (strong evidence)                                       | Typical / no judgement |
| CYP3A5 *3    | rs776746   | CT           | chip   | —             | —      | —     | CYP3A5: the T allele is the functional form (*1), less common; carriers metabolize a little faster some CYP3A substrates. (moderate evidence)           | Typical / no judgement |
| AGTR1 A1166C | rs5186     | AC           | chip   | —             | —      | —     | Angiotensin II AT1 receptor, the target of telmisartan; C allele is associated with more AT1 signal (telmisartan blocks this receptor). (weak evidence) | Typical / no judgement |

Favourable

- CYP3A4\*22 absent (rs35599367 GG): normal elimination of tadalafil

Unfavourable

- CYP3A5 \*1/\*3 (functional form present)

**Interpretation:** No variants that increase tadalafil exposure. Functional CYP3A5 plays a small role in tadalafil. The AGTR1 1166 AC (target of telmisartan) does not change the indication. The interaction to be monitored is pharmacological: strong inhibitors of CYP3A4 (clarithromycin, ketoconazole, grapefruit) increase tadalafil and rapamycin.

# Mechanisms of ageing

Each finding was linked to the aging mechanisms of the list of objectives for immortality (84 mechanisms). The table shows only the mechanisms played by this report.

| Mechanism                   | Unfavourable                             | Favourable                                             | Interpretation                                  |
|-----------------------------|------------------------------------------|--------------------------------------------------------|-------------------------------------------------|
| Vascular Aging              | NOS3 Glu298Asp and -786C in heterozygous | coronary metaGRS p5 (consensus p18); SIM1 without risk | Well, with nitric oxide just below the optimum. |
| Arterial stiffness          | NOS3 heterozygous                        | Telmisartan and tadalafil already taken                | Compensated by drugs.                           |
| Endocrine dysregulation     | —                                        | P89 genetic testosterone                               | Favourable.                                     |
| Insulin resistance          | GPS of diabetes p57                      | —                                                      | Medium; already treated in the plan.            |
| Intercellular Communication | —                                        | SIM1, PHF21B without risk                              | Favourable (central component).                 |

## Practical suggestions

All the suggestions below are suggestion (not applied): only from genetics, they are aligned with the v2 genetic plan and do not change the current stack without Sérgio F decision and the doctor.

| Topic                         | Suggestion (not applied)                                                                                                                                                                                                                                                      | Genetic basis                         |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| Aerobic and strength exercise | Regular aerobic exercise (e.g. 150-300 min/week) and strength training: is the most evident intervention for erectile function, because it increases the NOS3 in the endothelium by shear stress. <b>suggestion (not applied)</b>                                             | NOS3 Glu298Asp and -786C heterozygous |
| Diet nitrites                 | Plants rich in nitrate (arugula, spinach, beetroot, celery) every day; give a route of nitric oxide production that does not depend on NOS3. Avoid daily antiseptic mouthwash, which eliminates bacteria from the mouth that convert nitrate. <b>suggestion (not applied)</b> | NOS3 heterozygous                     |
| Interactions of tadalafil     | Avoid grapefruit and warn the physician before strong inhibitors of CYP3A4 (clarithromycin, itraconazole, ketoconazole), which increase tadalafil and rapamycin. Never associate nitrates (nitroglycerin) or poppers. <b>suggestion (not applied)</b>                         | Normal CYP3A4; CYP3A5 *1/*3           |
| Testosterone in analysis      | Include total testosterone, SHBG and free testosterone calculated in routine analyzes, because rapamycin can lower it. <b>suggestion (not applied)</b>                                                                                                                        | P89 genetic testosterone              |
| Sleep                         | 7-9 h sleep (nightly erections occur in REM sleep); melatonin 0.3-1 mg at night as in the plan. <b>suggestion (not applied)</b>                                                                                                                                               | Central function                      |
| Coordination with the plan    | LDL and ApoB as low as possible and high fiber; without alcohol. D3 10 000 IU and omega-3 3.2-4 g/day (already taking fish oil). Curcumin, luteolin, apigenin and quercetin in pause while ALT is high. <b>suggestion (not applied)</b>                                       | V2 plan rules                         |

## Limitations

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- The GWAS of erectile dysfunction are few and the score uses only 3 loci; the explained heritability is small.
- The NOS3 variants have small and heterogeneous effects between studies; the repetition of the 4 intron (4a/4b) is not read by the chip.
- The percentiles compare with the reference population (n=633).
- Only genes: does not use analysis or clinical evaluation.

## Methods and sources

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- Genotypes: Raw DNA of the chip (GRCh38, strand +), analyzed only locally; Missing SNPs (statistical imputation, public reference panel of complete 30x genomes), accepted with  $DR2 \geq 0.8$ .
- Public positions and alleles: Ensembl REST (only rsIDs). Locally calculated frequencies in the reference population ( $n=633$ ); effect alleles confirmed by the expected frequency.
- Scores: simple GWAS Catalog score (local copy) for erectile dysfunction; PGS000018 (coronary metaGRS) and PGS000014 (GPS of diabetes) of complete imputation; PGS003559 (testosterone) and PGS003552 (SHBG) reused from the andropause report.
- Main references: Jorgenson 2018 (SIM1 and erectile dysfunction); Bovijn 2019 (GWAS of erectile dysfunction and mendelian randomization); 2024 verma (biobank phenomen, PHF21B, SLC39A8, ZC3HC1); Safarinejad 2010 (NOS3 and erectile dysfunction); Forstermann 2012 (NOS3 and endothelial function); Gupta 2011 (life style and erectile dysfunction, meta-analysis); Kuehl 2001 (CYP3A5).
- Mapping for 84 aging mechanisms of the target list. Report generated locally (HTML → PDF).